

Day 5: Conditionals and Loops in Python

1■■■ Conditional Statements

Conditional statements allow decision-making in programs using **if**, **elif**, and **else**.

- 1 **if** – executes a block when a condition is true
- 2 **elif** – checks another condition if previous is false
- 3 **else** – runs if all previous conditions are false

2■■■ Loops in Python

Loops are used to repeat a block of code multiple times until a condition is met.

- 1 **for loop** – iterates over a sequence (like a list or range)
- 2 **while loop** – runs as long as a condition remains true

3■■■ Loop Control Statements

- 1 **break** – exits the loop immediately
- 2 **continue** – skips the rest of the loop iteration
- 3 **pass** – a null operation, does nothing

4■■■ Examples

```
# Even or Odd num = int(input("Enter a number: ")) if num % 2 == 0: print("Even")
else: print("Odd") # For loop example for i in range(1, 6): print(i) # While loop
countdown count = 5 while count > 0: print(count) count -= 1 print("Blast Off!")
```

5■■■ Mini Quiz Recap

Total Questions: 10 | Correct: 9 ■

Concept reinforced: condition checks, loop flow, control statements.

■ *Keep practicing! You're doing great on your 6-month AI & Python roadmap.*